

AWS Cloud Fundamentals & Security Notes

1. AWS IAM (Identity and Access Management)

What is IAM? / IAM Kya Hai?

AWS IAM is an access control service that manages **Who** can log into your AWS account and **What** resources they can use.

(IAM AWS ka security access system hai jo yeh decide karta hai ke kaun login kar sakta hai aur login hone ke baad kya kaam kar sakta hai.)

- **Authentication (Identity Check):** Verifies who you are using Username, Password, and MFA. (Aap ki pehchan verify karna.)
- **Authorization (Access Control):** Determines what actions you are allowed to perform. (Yeh decide karna ke aap ko kis cheez ki ijazat hai aur kis ki nahi.)

Core Components of IAM / IAM Ke Core Components:

Component	English Definition	Roman Urdu / Easy Explanation
IAM User	An identity created for a specific person or application.	Ek individual insaan ka account (Username & Password).
IAM Group	A collection of IAM Users sharing the same permissions.	Multiple users ka group (e.g., Developers, Admins).
IAM Policy	A JSON document that defines permissions (Allow or Deny).	Rules ki document jisme likha hota hai kya Allowed hai aur kya Denied.
IAM Role	Temporary access granted to users or AWS services.	Temporary access pass jo kisi system ya user ko temporary kaam ke liye milta hai.

Golden Rule: Principle of Least Privilege

Give users **only the minimum permissions** required to do their job—nothing more, nothing

(Users ko sirf utni hi permissions do jitni unke kaam ke liye zaroori hain—na us se zyada, na kam.)

AWS security is a shared effort between **AWS** and the **Customer**.

+-----+	
CUSTOMER RESPONSIBILITY	
(Security "IN" the Cloud)	
• Customer Data & Encryption	
• IAM & User Access Management	
• OS Patching & Application Code	
• Firewall & Network Rules (Security Groups)	
+-----+	
SPLIT RESPONSIBILITY	
v	
+-----+	
AWS RESPONSIBILITY	
(Security "OF" the Cloud)	
• Physical Data Centers & Building Security	
• Hardware Infrastructure (Servers, Drives)	
• Host OS & Virtualization (Hypervisors)	
• Network Infrastructure & Edge Locations	
+-----+	

As you move from **IaaS** → **PaaS** → **SaaS**, AWS takes on more operational burden, and customer responsibility decreases.

Service Model	Example	AWS Responsibility	Customer Responsibility
IaaS	AWS EC2	Physical Hardware, Facility, Network.	OS Patching, Firewall Rules, App Code, IAM, Data.

PaaS	AWS RDS	Physical Hardware, OS, DB Patching.	Application Code, DB Tables, IAM, Data.
SaaS	WorkMail, Gmail	Infrastructure, OS, Software & App Code.	Data and User Passwords/Access (IAM).

📌 **Key Rule:** Regardless of the service model, **Data Security** and **Access Control (IAM)** are ALWAYS the Customer's responsibility!

3. 🚀 The 6 Rs of Cloud Migration

The **6 Rs** represent six different strategies used when migrating applications and workloads from On-Premises to the Cloud.

1. Rehost (Lift and Shift)

- **Meaning:** Move applications to the cloud as-is, without making any changes.
- **Effort:** Very Low & Fast.
- **Analogy:** Moving furniture from an old house to a new house without changing anything.

2. Replatform (Lift, Tinker, and Shift)

- **Meaning:** Make small optimizations or minor upgrades (like switching to a managed DB) without changing the core application code.
- **Effort:** Moderate.
- **Analogy:** Moving a geyser to a new house and adding a smart automatic thermostat to it.

3. Repurchase (Drop and Shop)

- **Meaning:** Drop the old custom-built software completely and purchase a ready-to-use cloud product (SaaS).
- **Effort:** Fast deployment, but requires data migration.
- **Analogy:** Throwing away an old washing machine and subscribing to a laundry service.

4. Refactor / Re-architect

- **Meaning:** Completely rewrite the application code from scratch to leverage cloud-native features (Serverless, Microservices).
- **Effort:** High time and cost, but yields maximum efficiency and lowest long-term cost.
- **Analogy:** Rebuilding an old petrol car into a modern Electric Vehicle (EV).

5. Retire

- **Meaning:** Decommission or permanently turn off applications that are no longer useful.
- **Analogy:** Throwing away broken, unused junk before moving houses.

6. Retain (Revisit)

- **Meaning:** Keep certain applications or sensitive data on-premises for now and do not migrate them.
- **Analogy:** Leaving heavy family heirlooms at your hometown home rather than taking them to a rented apartment.